

MAT 426, Class 2: More on LU Decomposition

Review...

THEOREM *If A is a square matrix that can be reduced to row echelon form without using any row interchanges, then A has an LU factorization.*

An Easier way to find the LU Decomposition

If A can be reduced to row-echelon form _____ row interchanges, then the only row operation operation needed is _____

Note: It is VERY important that we perform the row reduction from _____ to _____ and _____ to _____.

Example 1. Continuing with Example 3 from Class 1, we found that

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

(a) Identify the row operations used (in the order applied).

(b) What do you notice about the multipliers in the row operations and the lower triangular matrix L ?

Example 2. Find the LU factorization of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 8 & 7 & 9 \end{bmatrix}$.

The General Case...

- The product of lower triangular matrices is lower triangular.
- A lower triangular matrix with nonzero diagonal entries is invertible and the inverse is lower triangular.
- Derivation of LU factorization.

ALGORITHM FOR LU FACTORIZATION

- (1) Reduce A to an _____ U by a sequence of _____ operations, if possible.
- (2) Place entries in L such that the _____ sequence of row operations reduce ____ to ____.

Example 3. Find an LU factorization of $\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$

Example 4. Solve $\begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$ using LU factorization.

Practical Consideration: Storage of L and U in Algorithm

How many flops are required for finding LU decomposition using Gaussian Elimination?

Note: Gaussian elimination can fail because a _____ can become _____.

Example 5. Consider the matrix $A = \begin{pmatrix} 0.0001 & 1 \\ 1 & 1 \end{pmatrix}$.

- (a) Use Gaussian elimination with three digit arithmetic to find the LU decomposition of A .

- (b) Use Gaussian elimination using three digit arithmetic but allowing a row interchange to find the LU decomposition of A .